Devixo Solutions

AI/ML Internship Program

Task 01

Python Programming, Data Analysis & Visualization

Dataset: Hotel Booking Demand (Kaggle)

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Internship Domain: AI/ML

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GitHub: github.com/Unees-Kiani/Task_01_AI_ML

1. Introduction

This report documents Task 01 of the Devixo Solutions AI/ML Internship Program. The task covers a complete data analysis workflow in Python: loading a real-world dataset, understanding its structure, cleaning and preprocessing it, performing exploratory and statistical analysis, creating visualizations, and drawing conclusions. The work was carried out in Google Colab using Pandas, NumPy, Matplotlib and Seaborn.

2. Objective

The objective is to build practical skills in data loading, cleaning, exploratory data analysis, statistics, visualization, insight generation and project documentation, as a foundation for later machine learning tasks.

3. Dataset Name

Hotel Booking Demand

4. Dataset Source

Kaggle, "Hotel Booking Demand" dataset (originally published by Antonio, Almeida and Nunes, 2019). The data was loaded into Google Colab with Pandas.

5. Dataset Description

The dataset contains booking records for two hotels, a City Hotel and a Resort Hotel. Each row is one booking and includes when the booking was made, when the guest arrived, how long they stayed, the number of guests, the market segment and customer type, the daily rate, special requests, and whether the booking was canceled.

6. Dataset Size

	Rows	Columns
Raw dataset	119,390	32
Cleaned dataset	87,303	33

The cleaned dataset has one more column than the raw data because two columns were added (total_guests and total_nights) and the company column was dropped.

7. Features / Columns Description

Column	Description
hotel	City Hotel or Resort Hotel
booking_canceled	1 if the booking was canceled, 0 if not
lead_time	Days between booking and arrival
arrival_year / month / week / day	Arrival date components
stays_in_weekend_nights / week_nights	Number of weekend and weekday nights booked
adults, children, babies	Number of guests of each type
meal	Meal plan booked

Column	Description
country	Guest country of origin
market_segment	How the booking was sourced (Online TA, Direct, Corporate, etc.)
customer_type	Transient, Transient-Party, Contract or Group
average_daily_rate	Average price per night (renamed from adr)
previous_cancellations	Past bookings canceled by the guest
required_car_parking_spaces	Parking spaces requested
total_of_special_requests	Number of special requests made
total_guests, total_nights	New columns: adults + children + babies; weekend + week nights

8. Methodology

The project followed the standard workflow: dataset selection, data loading, data understanding, data cleaning, exploratory data analysis, statistical analysis, data visualization, insights and conclusion, then documentation and GitHub organization.

9. Data Loading

The libraries Pandas, NumPy, Matplotlib and Seaborn were imported and the dataset was read into a Pandas DataFrame. The first five records, last five records and a random sample were displayed, followed by the shape, column names, data types and non-null counts. Columns were classified as numerical, categorical and date/time. Numerical columns are used for statistics and correlation, categorical columns for grouping and counts, and the date column for time-based analysis.

10. Data Cleaning

• **Missing values:** The company column was about 94% empty, so it was dropped. Missing agent values were filled with 0, because a blank means no agent was used. The few missing children values were filled with the median, and missing country values with the mode, since country is categorical.

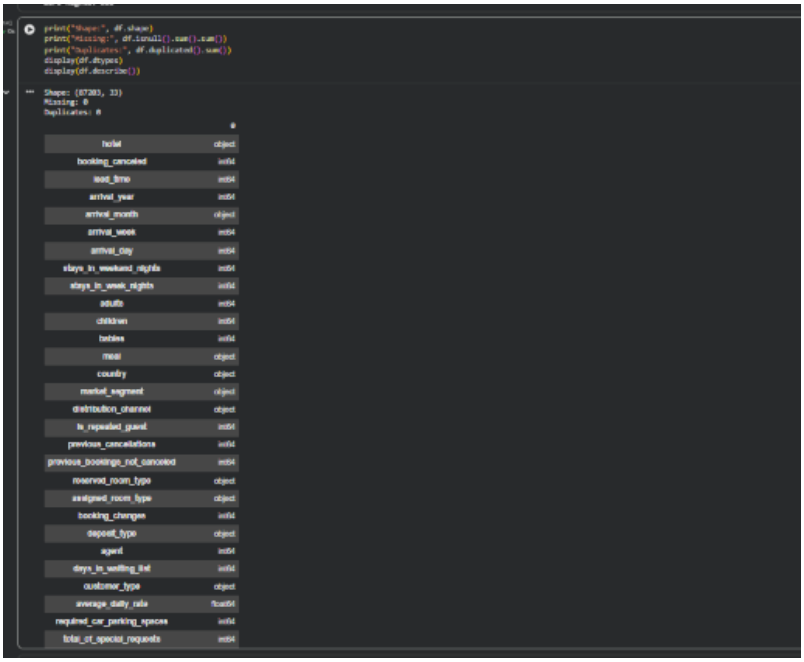
• **Duplicates:** Duplicate records were counted and removed, and it was verified that none remained.

• **Data types:** reservation_status_date was converted from text to datetime, and agent and children were converted to integers.

• **Column names:** Unclear names were changed, for example adr to average_daily_rate and is_canceled to booking_canceled.

• **Invalid values:** Bookings with zero guests or a negative price are logically impossible and were removed. Zero-night bookings were kept because they can be same-day stays or no-shows. Extreme values (for example a daily rate of 5,400) were identified as outliers. For price charts, rates above 500 were filtered so the charts stay readable, while the values remain in the dataset.

• **Verification:** After cleaning, the dataset has 87,303 rows, 0 missing values and 0 duplicates.

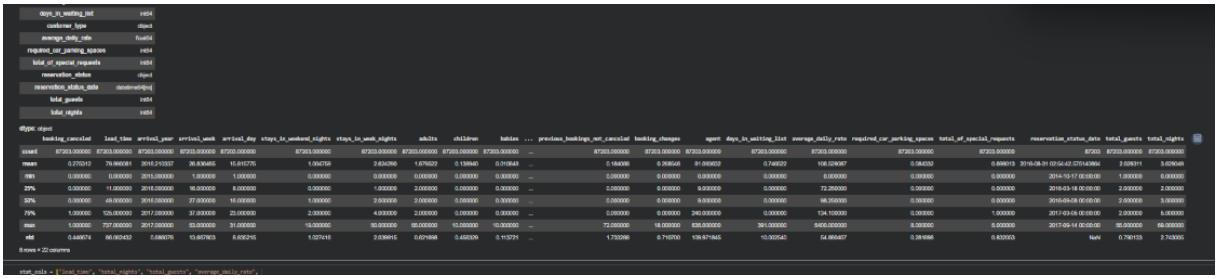


```
print("Shape:", df.shape)
print("Missing:", df.isnull().sum().sum())
print("Duplicates:", df.duplicated().sum())
display(df.dtypes)
display(df.describe())
```

```
Shape: (87203, 33)
Missing: 0
Duplicates: 0
```

	0
hotel	object
booking_canceled	int64
lead_time	int64
arrival_year	int64
arrival_month	object
arrival_week	int64
arrival_day	int64
stays_in_weekend_nights	int64
stays_in_week_nights	int64
adults	int64
children	int64
babies	int64
meal	object
country	object
market_segment	object
distribution_channel	object
is_repeated_guest	int64
previous_cancellations	int64
previous_bookings_not_canceled	int64
reserved_room_type	object
assigned_room_type	object
booking_changes	int64
deposit_type	object
agent	int64
days_in_waiting_list	int64
customer_type	object
average_daily_rate	float64
required_car_parking_spaces	int64
total_of_special_requests	int64

Figure A: Verification of the cleaned dataset (shape, missing values, duplicates, data types)



	days_in_waiting_list	customer_type	average_daily_rate	required_car_parking_spaces	total_of_special_requests	reservation_status	reservation_status_date	total_guests	total_nights
dtype:	int64	object	float64	int64	int64	object	datetime64[ns]	int64	int64

	booking_canceled	lead_time	arrival_year	arrival_week	arrival_day	stays_in_weekend_nights	stays_in_week_nights	adults	children	babies	...	previous_bookings_not_canceled	booking_changes	agent	days_in_waiting_list	average_daily_rate	required_car_parking_spaces	total_of_special_requests	reservation_status_date	total_guests	total_nights
count	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	...	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203.000000	87203	87203.000000	87203.000000
mean	0.275212	79.960581	2016.210337	26.836455	15.815775	1.004758	2.624268	1.876522	0.139540	0.010548	...	0.194098	0.358548	81.869052	0.746822	106.529087	0.084232	0.699013	2016-08-31 00:54:42.070143964	2.029311	3.629349
min	0.000000	0.000000	2015.000000	1.000000	1.000000	0.000000	0.000000	0.000000	0.000000	0.000000	...	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	2014-10-17 00:00:00	1.000000	0.000000
25%	0.000000	11.000000	2016.000000	16.000000	8.000000	0.000000	1.000000	2.000000	0.000000	0.000000	...	0.000000	0.000000	9.000000	0.000000	72.250000	0.000000	0.000000	2016-03-18 00:00:00	2.000000	2.000000
50%	0.000000	49.000000	2016.000000	27.000000	16.000000	1.000000	2.000000	2.000000	0.000000	0.000000	...	0.000000	0.000000	9.000000	0.000000	98.350000	0.000000	0.000000	2016-09-08 00:00:00	2.000000	3.000000
75%	1.000000	125.000000	2017.000000	37.000000	23.000000	2.000000	4.000000	2.000000	0.000000	0.000000	...	0.000000	0.000000	240.000000	0.000000	134.100000	0.000000	1.000000	2017-03-06 00:00:00	2.000000	5.000000
max	1.000000	737.000000	2017.000000	53.000000	31.000000	19.000000	50.000000	55.000000	10.000000	10.000000	...	72.000000	18.000000	535.000000	391.000000	5400.000000	8.000000	5.000000	2017-09-14 00:00:00	55.000000	69.000000
std	0.446874	86.062432	0.680076	13.667803	8.836215	1.037416	2.039815	0.621698	0.468329	0.113721	...	1.733098	0.710700	159.671465	10.000540	54.866407	0.281956	0.832953	NaN	0.790133	2.743005

8 rows × 22 columns

```
stat_cols = ['lead_time', 'total_nights', 'total_guests', 'average_daily_rate',]
```

Figure B: Basic statistics of the cleaned dataset (describe)

11. Exploratory Data Analysis

EDA examined which variables matter, which categories are most common, where the highest and lowest values are, whether outliers exist, and how variables relate. The main questions were about cancellations, seasonality, pricing and booking channels.

12. Statistical Analysis

Mean, median, mode, minimum, maximum, range, standard deviation and variance were calculated for key numerical columns, and a correlation matrix was created.

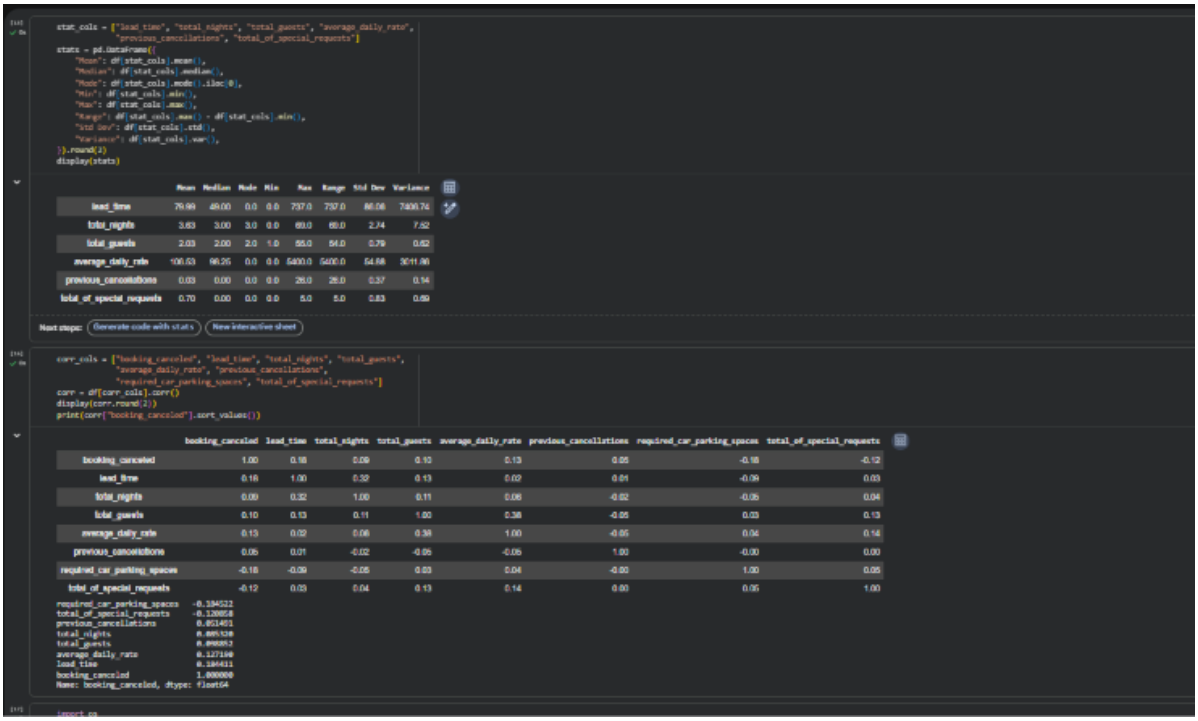


Figure C: Code and output for descriptive statistics and the correlation matrix

Key statistical results

• **Lead time:** The mean is about 80 days and the median is 49. A few very early bookings (maximum 737 days) pull the mean up, so the median is more typical. The standard deviation (about 86) is larger than the mean, so lead times vary a lot.

• **Daily rate:** The mean is 106.53 and the median is 98.20. The maximum of 5,400 is an extreme outlier. The mode of 0 comes from zero-rate bookings and is not a typical price.

• **Stay length:** Mean 3.63, median 3, mode 3, so most guests stay about three nights.

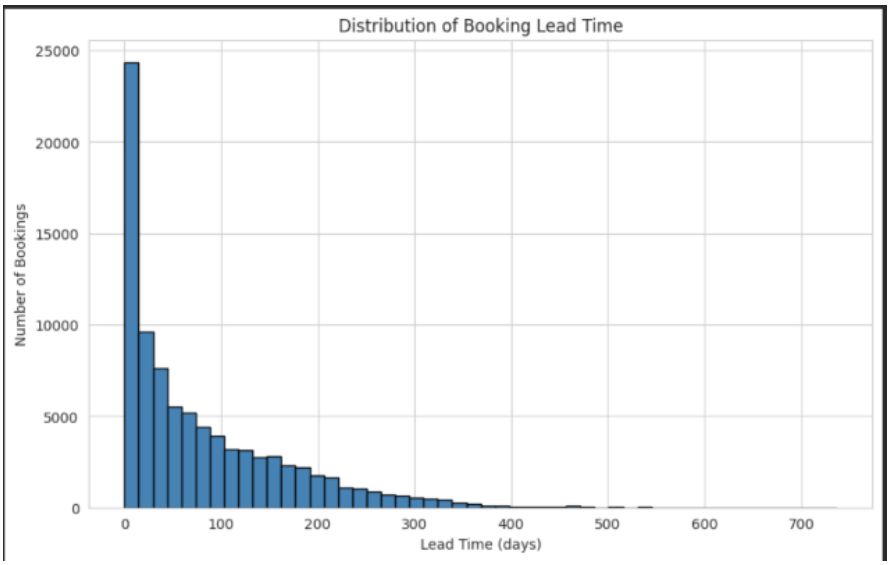
• **Guests:** Mean, median and mode are all around 2 with a low standard deviation (0.79), so most bookings are for two people.

• **Correlation with cancellation:** Lead time is the strongest positive correlate (+0.18), and required parking spaces is the strongest negative (-0.18), followed by special requests (-0.12). All correlations are weak to moderate, which is normal for real-world data.

13. Data Visualizations

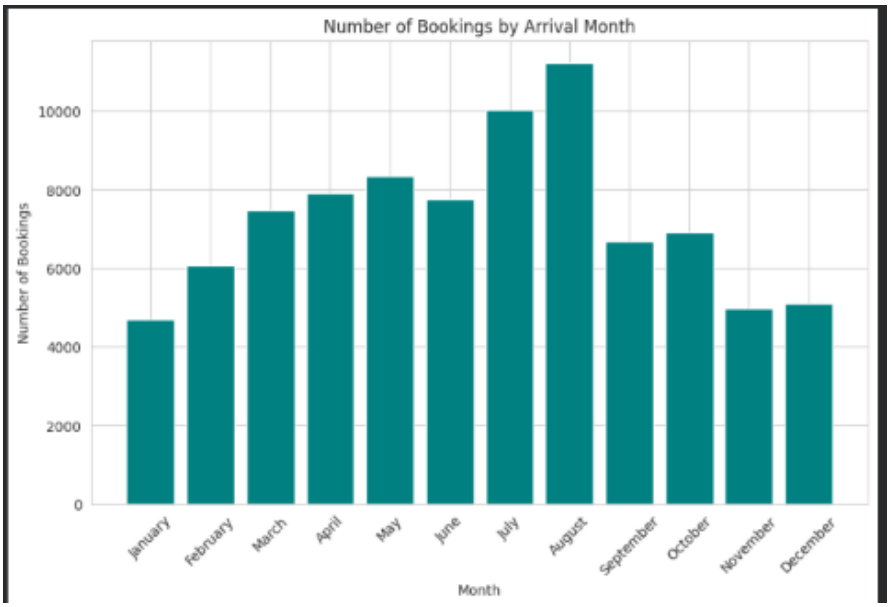
Ten visualizations were created. Each has a title, axis labels and a suitable figure size, and each is explained below.

Visualization 1. Histogram: Distribution of Booking Lead Time



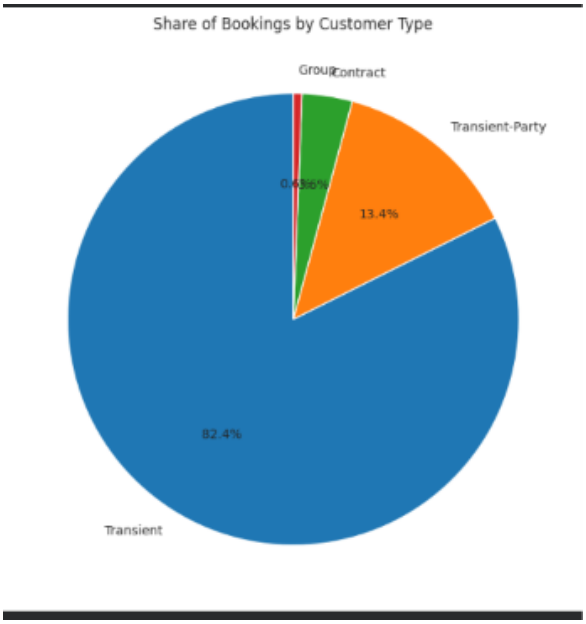
The distribution is heavily right-skewed. About 24,000 bookings were made within roughly two weeks of arrival, and counts fall steadily after that. Most guests book late, but a small group books more than 300 days ahead, which is why the mean is far above the median.

Visualization 2. Bar Chart: Bookings by Arrival Month



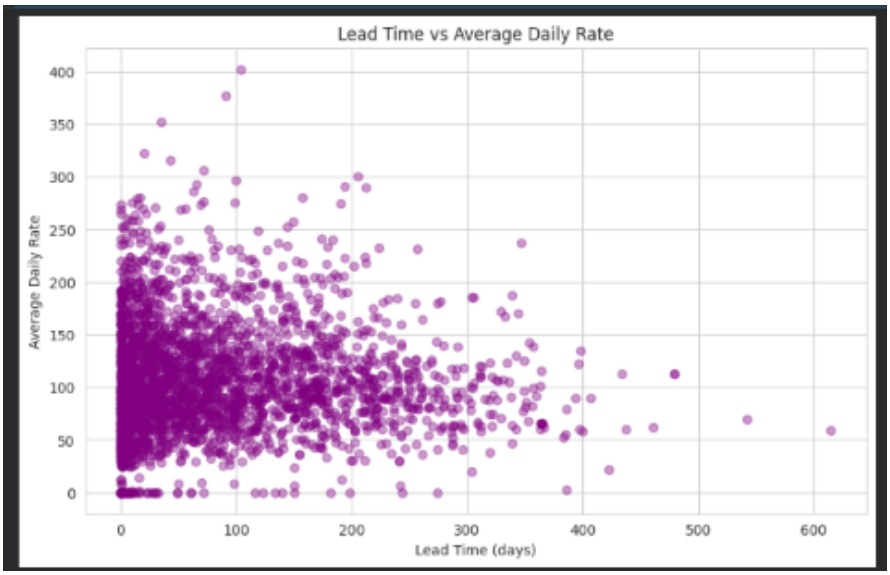
August is the busiest month (about 11,000 bookings), followed by July. January is the quietest (about 4,700). Demand peaks in summer, which matters for staffing and pricing.

Visualization 3. Pie Chart: Share of Bookings by Customer Type



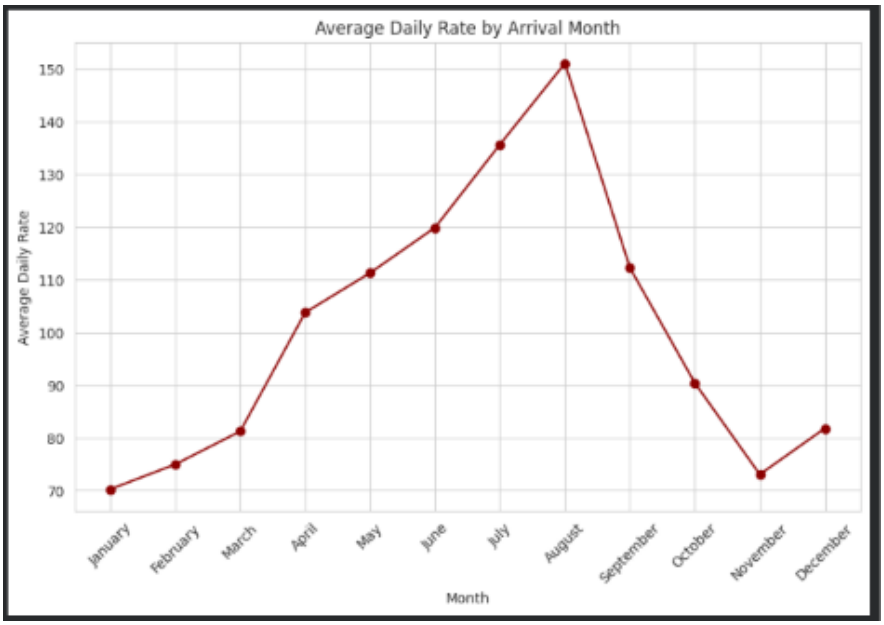
Transient customers make up 82.4% of bookings and Transient-Party 13.4%. Contract and Group bookings together are under 5%, so the hotels depend mainly on individual travelers.

Visualization 4. Scatter Plot: Lead Time vs Average Daily Rate



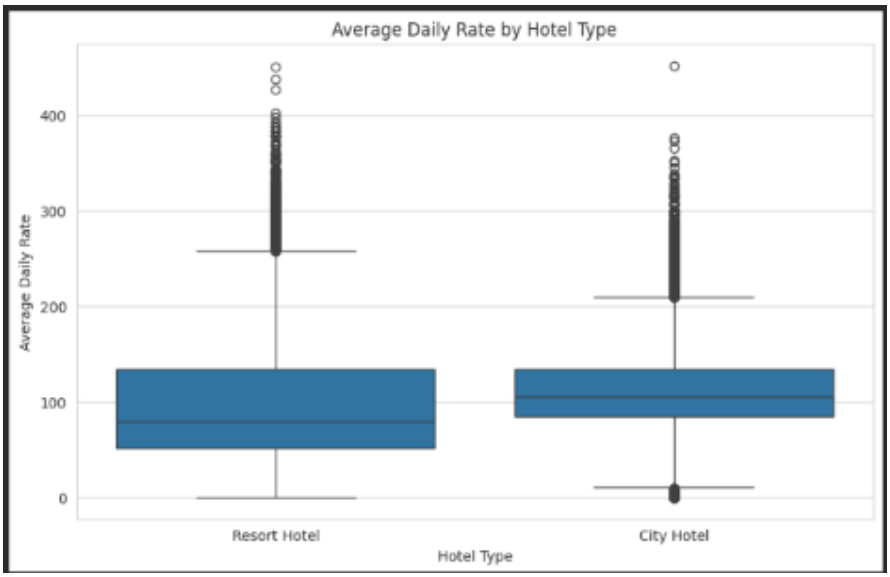
The points form a wide cloud with no clear trend. Booking earlier does not mean paying more or less, which matches the very weak correlation of about 0.02.

Visualization 5. Line Chart: Average Daily Rate by Arrival Month



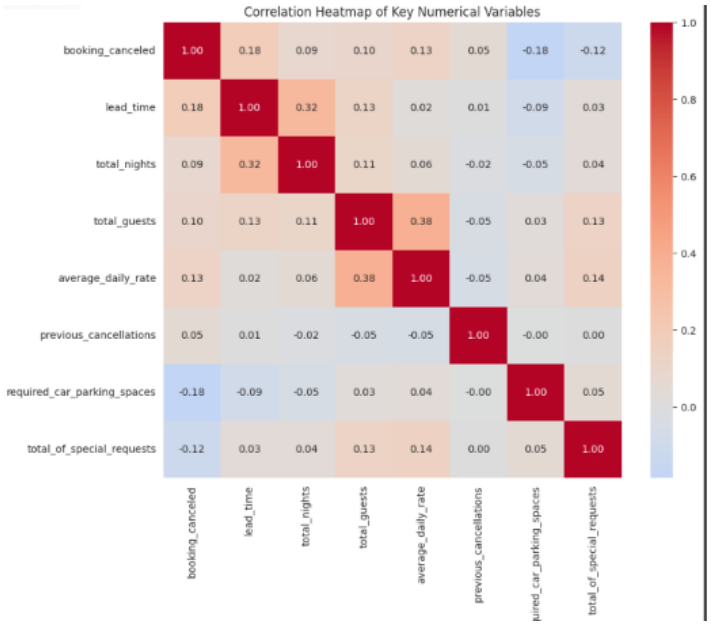
The average rate rises from about 70 in January to a peak of about 150 in August, then drops sharply to about 73 in November. Prices follow demand, and the busiest month is also the most expensive.

Visualization 6. Box Plot: Average Daily Rate by Hotel Type



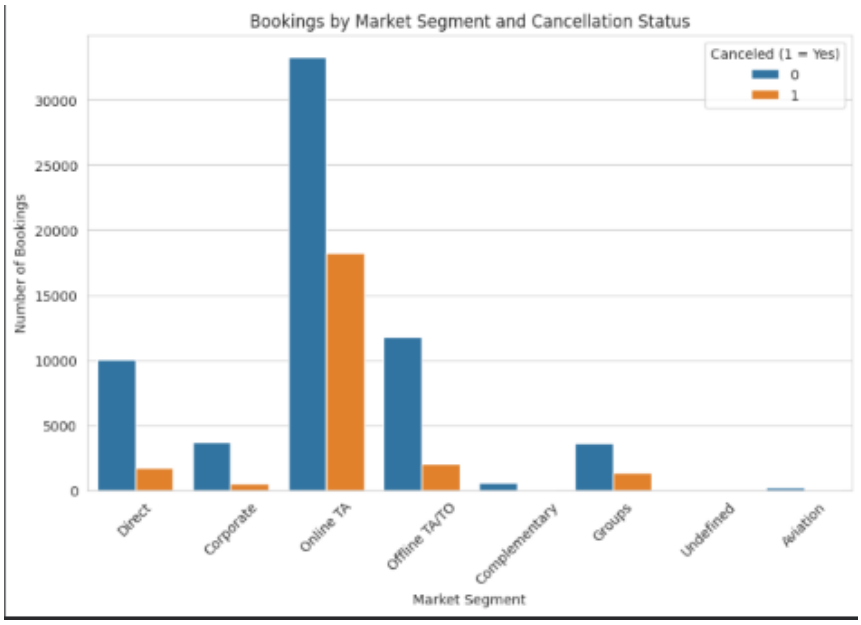
City Hotel has a higher median rate (about 105) than Resort Hotel (about 80). Resort Hotel has a wider spread and more high-price outliers, probably from peak-season rates.

Visualization 7. Correlation Heatmap of Key Numerical Variables



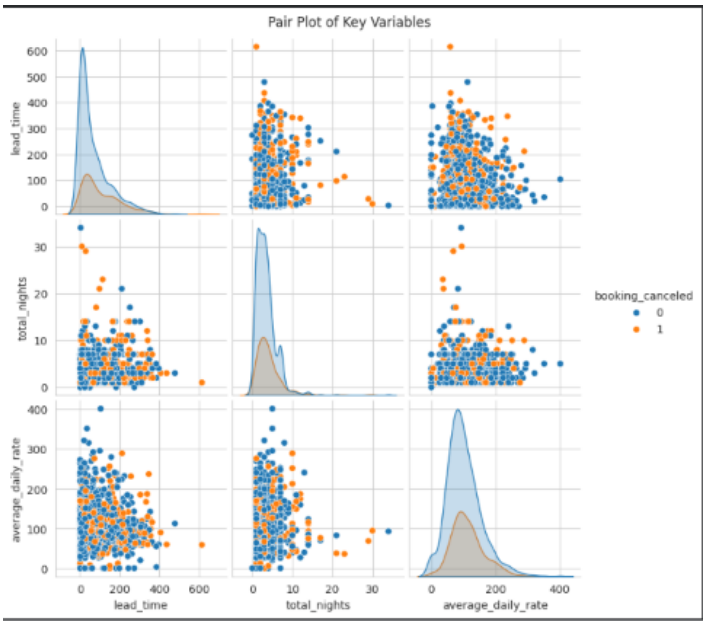
The strongest correlations with cancellation are lead time (+0.18) and parking spaces (-0.18). Total guests and daily rate are correlated at +0.38, and lead time and total nights at +0.32.

Visualization 8. Count Plot: Bookings by Market Segment and Cancellation Status



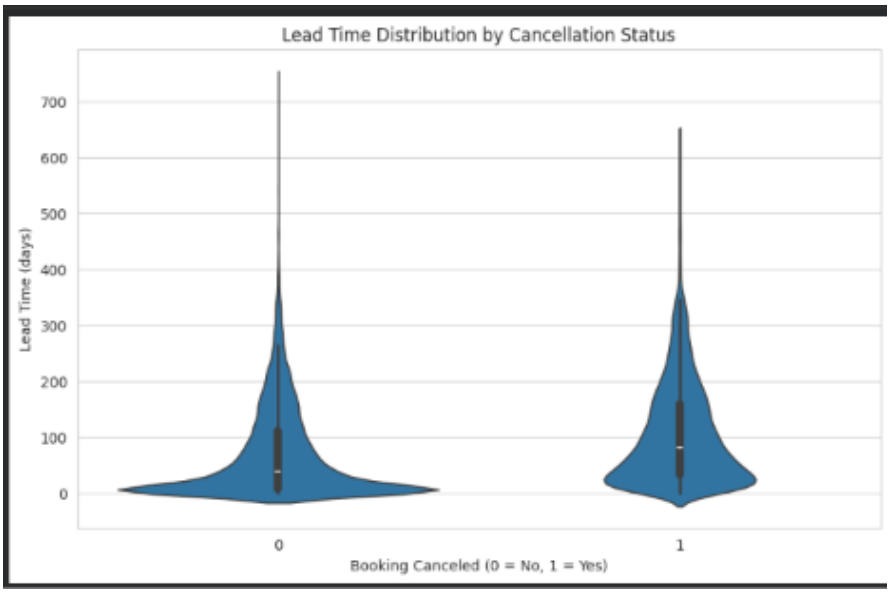
Online travel agency (TA) bookings are by far the most numerous and also have the most cancellations. Corporate and Direct bookings are mostly kept.

Visualization 9. Pair Plot of Key Variables



Canceled bookings (orange) are spread toward longer lead times, while kept bookings (blue) cluster near zero lead time. Total nights and daily rate show no strong separation.

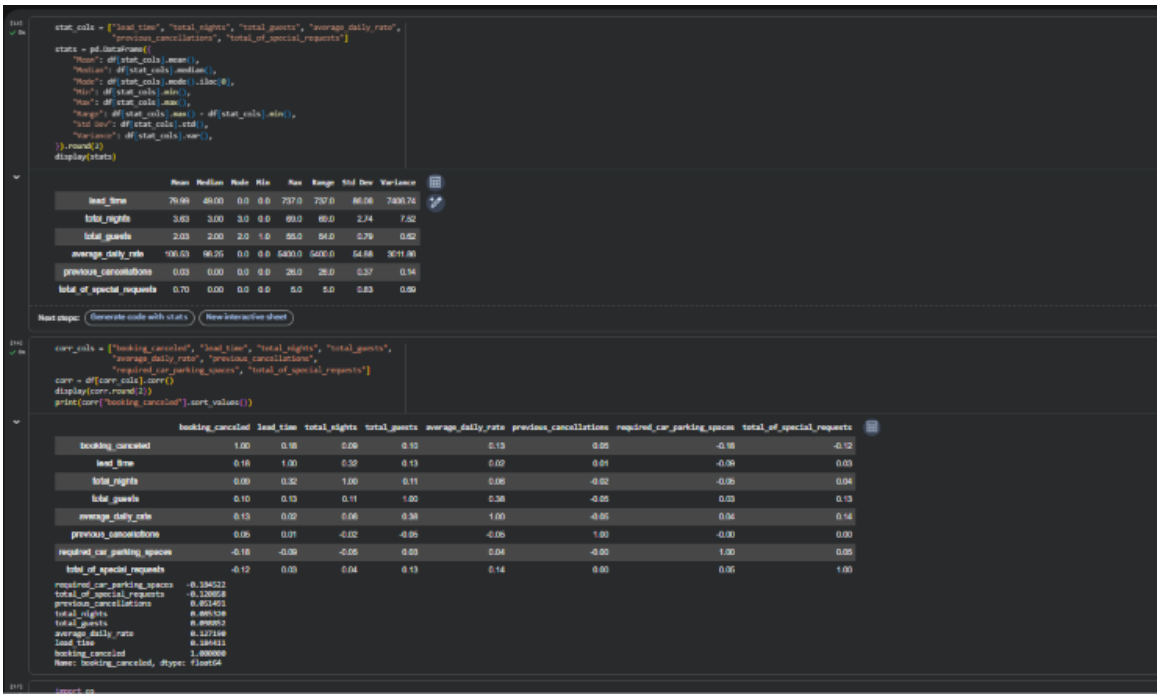
Visualization 10. Violin Plot: Lead Time by Cancellation Status



Non-canceled bookings are concentrated at short lead times. Canceled bookings are spread wider, with a median of about 80 days versus about 40, confirming that longer lead time goes with more cancellations.

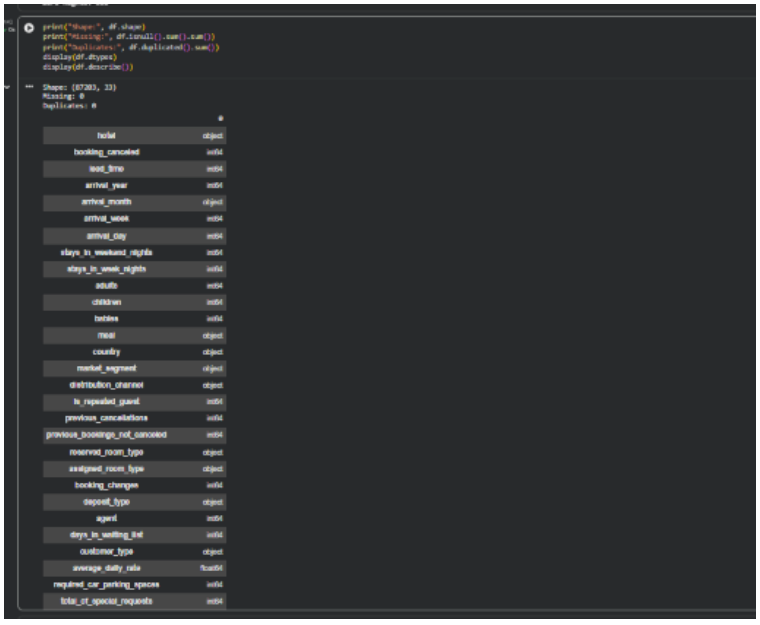
14. Code Screenshots

The screenshots below show the analysis code running in Google Colab (statistics and correlation, and the extra summary statistics used for the insights).

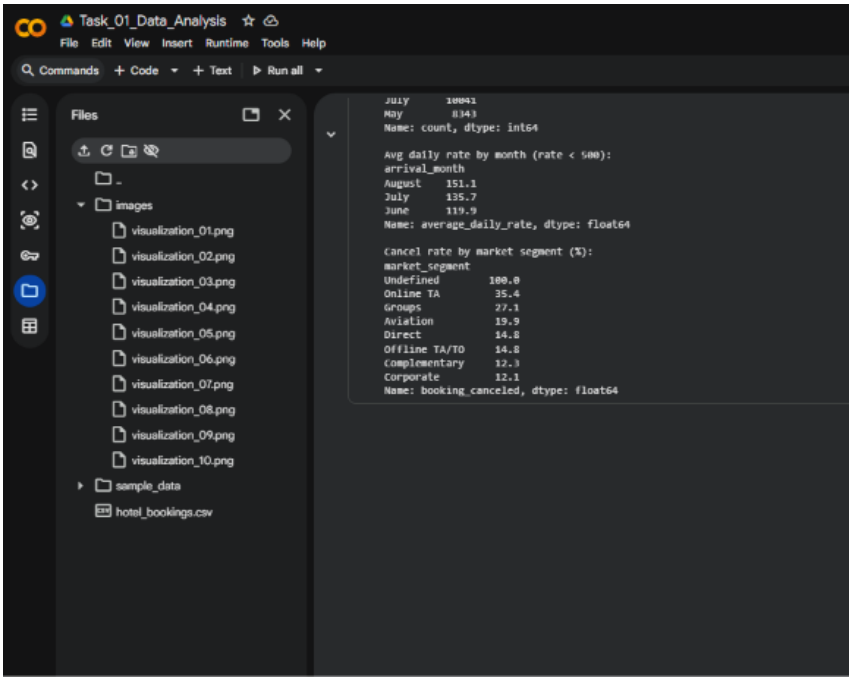


Code for descriptive statistics and the correlation matrix

15. Output Screenshots



Output: cleaned dataset verification



Output: average daily rate by month and cancellation rate by market segment

16. Graph Screenshots

All ten graph screenshots are included in Section 13. The image files are also saved in the images folder of the GitHub repository (visualization_01.png to visualization_10.png).

17. Major Findings

• About 27% of bookings were canceled.

• Lead time is the strongest positive correlate of cancellation, so bookings made far in advance are riskier.

• Summer is the peak season: August is both the busiest and the most expensive month.

• Online travel agency bookings have a much higher cancellation rate than Corporate or Direct bookings.

• Transient customers make up more than 80% of all bookings.

18. Insights (at least 10)

1. After cleaning, the dataset has 87,303 bookings and 33 columns, with no missing values and no duplicates.

2. About 27% of bookings were canceled.

3. Lead time is right-skewed: the mean is about 80 days and the median is 49.

4. The most common booking is made within two weeks of arrival.

5. August is the busiest month and January is the quietest.

6. Average daily rate peaks in August (about 151) and is lowest in winter, at about 70 in January.

7. Transient customers make up 82.4% of all bookings.

8. Online TA has the highest cancellation rate among large market segments (35.4%), compared with about 12% for Corporate and 15% for Direct bookings.

9. Longer lead time is the strongest positive correlate of cancellation (+0.18), and required parking spaces is the strongest negative (-0.18).

10. Total guests and daily rate are positively correlated (+0.38), so larger parties pay higher rates.

11. Daily rate has extreme outliers, with a maximum of 5,400.

12. The Undefined market segment shows 100% cancellation but has very few records, so it is not reliable.

13. The typical stay is three nights, and most bookings are for two guests.

19. Conclusion

For this task, I selected the Hotel Booking Demand dataset from Kaggle, which contains over 119,000 booking records from a City Hotel and a Resort Hotel. The purpose of the analysis was to understand booking behaviour, pricing patterns, and the factors linked to cancellations. During cleaning, I dropped the company column because about 94% of its values were missing, filled the remaining missing values using the median, mode, or a meaningful default, removed duplicate records, corrected data types, renamed unclear columns, and removed impossible records such as bookings with zero guests. This left 87,303 clean records. I then calculated descriptive statistics (mean, median, mode, range, standard deviation, variance), built a correlation matrix, and created ten visualizations, including a histogram, box plot, heatmap, pair plot, and violin plot.

The most important findings were that about 27% of bookings were canceled, that longer lead times are the strongest predictor of cancellation, that August is both the busiest and the most expensive

month, and that Online Travel Agency bookings are canceled far more often than Corporate or Direct bookings. Through this project I learned how to clean real-world data, choose suitable methods for missing values, interpret statistics and charts, and explain results clearly. These findings could help a hotel plan pricing, manage overbooking risk, and focus on lower-cancellation booking channels.

20. GitHub Repository Link

https://github.com/Unees-Kiani/Task_01_AI_ML